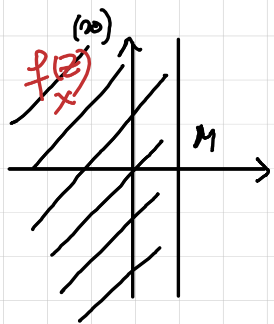
Seminar 11

① $f \in \mathcal{H}(\mathbb{C})$

a) $\exists M \in \mathbb{R}$ a.î. $\operatorname{Re} f(z) \le M$, $\forall z \in \mathbb{C}$ $\Big\} \Rightarrow f \equiv c$



T. LIOUVILLE O funcție întreagă și mărginită este constantă

$$\left[\begin{array}{l} f \in \mathcal{H}(\mathbb{C}) \\ \exists M \ge 0 \text{ cu } |f(z)| \le M, \forall z \in \mathbb{C} \end{array}\right\} \Rightarrow f \equiv c \Big]$$

$e^z = e^x(\cos y + i \sin y)$, $z = x+iy$, $x = \operatorname{Re} z$

$|e^z| = e^x = e^{\operatorname{Re} z}$

$|e^{f(z)}| = e^{\operatorname{Re} f(z)}$

Definim $g(z) = e^{f(z)}$, $g: \mathbb{C} \to \mathbb{C}$

compunere de funcții întregi $\Rightarrow g \in \mathcal{H}(\mathbb{C})$

$|g(z)| = |e^{f(z)}| \le e^M$, $e^M > 0$ $\Big\} \overset{\text{T. Liouville}}{\Rightarrow}$ g constantă

e^f constantă $\Rightarrow$ $g'(z) = 0$, $\forall z \in \mathbb{C}$

$e^{z+2k\pi i} = e^z$

$g'(z) = (e^{f(z)})' = \underbrace{e^{f(z)}}_{\in \mathbb{C}^*} \cdot f'(z) = 0 \Rightarrow f'(z) = 0$, $\forall z \in \mathbb{C}$

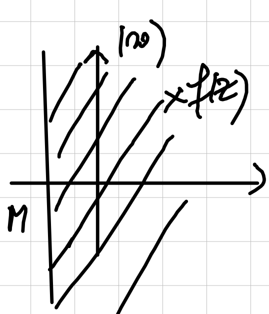
Curs 5

$\mathbb{C}$ domeniu (deschisă + conexă) $\Big\} \Rightarrow f(z) = c$, $\forall z \in \mathbb{C}$

$f \equiv c$

b) $f \in \mathcal{H}(\mathbb{C})$

$\exists M \in \mathbb{R}$ cu $\operatorname{Re} f(z) \ge M$ $\quad | \cdot (-1)$



$-\operatorname{Re} f(z) \le -M \iff \operatorname{Re}(-f(z)) \le -M \overset{a)}{\underset{-f}{\Rightarrow}} -f \equiv c$

$f \equiv -c$

② $\nexists f: \mathbb{C} \to U$, $f \in \mathcal{H}(\mathbb{C})$, f bijectivă, $U = U(0;1)$ discul unitate

pp $\exists f: \mathbb{C} \to U$, $f \in \mathcal{H}(\mathbb{C})$, f bijectivă

$$\Downarrow$$

$$\left.\begin{array}{l} \forall z \in \mathbb{C} \Rightarrow f(z) \in U \\ U = \{z \in \mathbb{C} : |z| < 1\} \end{array}\right\} \Rightarrow |f(z)| < 1, \ \forall z \in \mathbb{C} \Rightarrow \left.\begin{array}{l} f \text{ mărg} \\ f \in \mathcal{H}(\mathbb{C}) \end{array}\right\} \overset{\text{T. Liouville}}{\Rightarrow} f \equiv c$$

$f(z) = c$, $\forall z \in \mathbb{C}$ $\Rightarrow$ f nu este injectivă

$\lightning$ f bijectivă

$f: \mathbb{C} \to U$ $\qquad \mathbb{C} = \mathbb{R}^2$, $U = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$

$$z = x + iy$$

f de clasă C^∞ pe $\mathbb{C}$ $\Leftrightarrow$ $u = \operatorname{Re} f$, $v = \operatorname{Im} f$ de clasă C^∞ pe $\mathbb{C}$

f bijectivă (neconstantă)

$$f(z) = \frac{z}{1+|z|} \qquad |f(z)| = \frac{|z|}{1+|z|} < 1 \Leftrightarrow |z| < 1 + |z| \Leftrightarrow 0 < 1 \text{ „A"}$$

$$\Rightarrow f(z) \in U, \ \forall z \in \mathbb{C}$$

$$z = x + iy \Rightarrow u(z) = \frac{x}{1+\sqrt{x^2+y^2}}, \quad v(z) = \frac{y}{1+\sqrt{x^2+y^2}} \quad C^\infty(\mathbb{C}) \text{ real}$$

$$f(z_1) = f(z_2) \Rightarrow \frac{z_1}{1+|z_1|} = \frac{z_2}{1+|z_2|} \Rightarrow \begin{cases} \dfrac{|z_1|}{1+|z_1|} = \dfrac{|z_2|}{1+|z_2|} & (1) \\ \arg \dfrac{z_1}{1+|z_1|} = \arg \dfrac{z_2}{1+|z_2|} & (2) \end{cases}$$

$$(1) \Rightarrow |z_1| + \cancel{|z_1||z_2|} = |z_2| + \cancel{|z_1||z_2|} \Rightarrow |z_1| = |z_2|$$

$$\arg(\underbrace{1+|z_1|}_{\in \mathbb{R}_+^*}) = \arg(\underbrace{1+|z_2|}_{\in \mathbb{R}_+^*}) = 0 \qquad \arg \frac{w_1}{w_2} = \arg w_1 - \underbrace{\arg w_2}_{=0}$$

$$(2) \Rightarrow \left.\begin{array}{l} \arg z_1 = \arg z_2 \\ |z_1| = |z_2| \end{array}\right\} \Rightarrow z_1 = z_2 \Rightarrow f \text{ injectivă}$$

Fie $w \in U$, $\exists ?\, z \in \mathbb{C}$ cu $f(z) = w \Leftrightarrow \dfrac{z}{1+|z|} = w \Rightarrow \begin{cases} \dfrac{|z|}{1+|z|} = |w| \\ \arg \dfrac{z}{1+|z|} = \arg w \end{cases}$

$$\Updownarrow$$

$$|w| < 1$$

$$\Updownarrow$$

$$1 - |w| > 0$$

$$\Rightarrow \begin{cases} |z| = |w| + |z| \cdot |w| \\ \arg z = \arg w \end{cases} \Rightarrow \begin{cases} |z| = \dfrac{|w|}{1-|w|} \\ \arg z = \arg w \end{cases}$$

$$\underbrace{|z| e^{i \arg z}}_{z} = \frac{|w|}{1-|w|} e^{i \arg w} \Rightarrow z = \frac{w}{1-|w|} \in \mathbb{C}$$

$\Rightarrow f$ surjectivă $\qquad f(\mathbb{C}) = U$

$$\sum_{n=0}^{\infty} a_n(z-z_0)^n = a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \dots + a_n(z-z_0)^n + \dots$$

serie de puteri $\qquad a_n \in \mathbb{C}$ coeficienții seriei de puteri

T. Cauchy-Hadamard

$\sum_{n=0}^{\infty} a_n(z-z_0)^n$, $\frac{1}{R} = \overline{\lim_{n\to\infty}} \sqrt[n]{|a_n|}$, R = rază de convergență

a) seria de puteri este convergentă abs. și uniform pe compacte în $U(z_0;R)$

$U(z_0;R)$ disc de convergență

b) seria de puteri divergentă pe $\mathbb{C} \setminus \overline{U}(z_0;R)$

c) $S(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n$, $z \in U(z_0;R)$; $S \in \mathcal{H}(U(z_0;R))$

d) seria derivată $\sum_{n=1}^{\infty} n a_n (z-z_0)^{n-1}$ convergentă abs + unif pe compacte pe $U(z_0;R)$

are suma $= S'(z)$

3) a) $\sum_{n=0}^{\infty} [3+(-1)^n]^n z^n$ $\qquad$ discul de convergență este $U(0;\frac{1}{4})$

$z_0 = 0$, $a_n = [3+(-1)^n]^n \in \mathbb{R}_+^*$

$k \in \mathbb{N}$ $\quad a_{2k} = 4^{2k}$ $\quad \sqrt[2k]{|a_{2k}|} = 4 \to 4 \ (n\to\infty)$

$a_{2k+1} = 2^{2k+1}$ $\quad \sqrt[2k+1]{|a_{2k+1}|} = 2 \to 2 \ (n\to\infty)$

$\overline{\lim_{n\to\infty}} \sqrt[n]{|a_n|} = \max\{2,4\} = 4 \Rightarrow \frac{1}{R} = 4 \Rightarrow \boxed{R = \frac{1}{4}}$

b) $\sum_{n=0}^{\infty} z^{2^n} = z + z^2 + z^4 + z^8 + \dots$

$\sum_{n=0}^{\infty} a_n(z-z_0)^n = \sum_{n=0}^{\infty} 1\cdot(z-0)^n = \sum_{n=0}^{\infty} z^n$

$z_0 = 0$, $a_n = 1$

$z_0 = 0$, $a_n = \begin{cases} 1, & n = 2^k \\ 0, & n \neq 2^k \end{cases}$

$(0\cdot z^3, \ 0\cdot z^5)$

$\sqrt[n]{|a_n|} = \begin{cases} 1, & n = 2^k \\ 0, & n \neq 2^k \end{cases} \Rightarrow \frac{1}{R} = \max\{0,1\} = 1 \Rightarrow R = 1$

$U(0;1)$ discul de convergență

d) $\exists \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = l \xRightarrow{\text{Cauchy}} \lim_{n\to\infty} \sqrt[n]{|a_n|} = l$

$R = \lim_{n\to\infty} \frac{|a_n|}{|a_{n+1}|}$

$\sum_{n=1}^{\infty} \frac{n!}{n^n}(z-i)^n$ $\qquad a_n = \frac{n!}{n^n}$ $\qquad a_n > 0, \forall n \in \mathbb{N}^*$

$R = \lim_{n\to\infty} \frac{a_n}{a_{n+1}} = \lim_{n\to\infty} \frac{\cancel{n!}}{n^n} \cdot \frac{(n+1)^{\cancel{n+1}}}{\cancel{(n+1)}!} = \lim_{n\to\infty} \left(\frac{n+1}{n}\right)^n = \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e$

$U(i;e)$ discul de convergență

Teorema dezvoltării în serie Taylor

$f \in \mathcal{H}(U(z_0;r)) \Rightarrow \exists! \sum_{n=0}^{\infty} a_n(z-z_0)^n$ cu $R \geq r$ a.î.

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n \quad , \forall z \in U(z_0;r)$$

$$n \in \mathbb{N} \quad , \boxed{a_n = \frac{f^{(n)}(z_0)}{n!}} = \frac{1}{2\pi i}\int_{\gamma} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta \quad , \gamma = \partial U(z_0;\rho)$$

$$0 < \rho < r$$

4b) $f : \mathbb{C} \to \mathbb{C} \ , f(z) = e^z \quad \Rightarrow f^{(n)}(z) = e^z \ , \forall n \in \mathbb{N}, \forall z \in \mathbb{C}$

$$z_0 = 0 \Rightarrow f^{(n)}(0) = e^0 = 1$$

$$a_n = \frac{f^{(n)}(0)}{n!} = \frac{1}{n!}$$

T. dezv Taylor

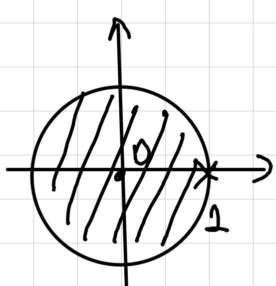
$$\boxed{e^z = \sum_{n=0}^{\infty} \frac{1}{n!} z^n \quad , z \in \mathbb{C}}$$

$$R = \lim_{n\to\infty} \frac{a_n}{a_{n+1}} = \lim_{n\to\infty} \frac{1}{n!} \cdot \frac{(n+1)!}{1} = \lim_{n\to\infty} (n+1) = \infty$$

4a) $f : \mathbb{C} \setminus \{1\} \to \mathbb{C} \quad , f(z) = \frac{1}{1-z}$

$$f'(z) = \frac{1}{(1-z)^2} \quad , f''(z) = \frac{1\cdot 2}{(1-z)^3} \ , \ldots , f^{(n)}(z) = \frac{n!}{(1-z)^{n+1}}$$

$$z_0 = 0 \quad \Rightarrow f^{(n)}(0) = n! \quad \Rightarrow a_n = \frac{f^{(n)}(0)}{n!} = \frac{n!}{n!} = 1$$



$$f \in \mathcal{H}(U(0;1)) \overset{\text{T dezv. Taylor}}{\Longrightarrow}$$

$$\boxed{\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n \qquad z \in U(0;1) \Leftrightarrow |z| < 1}$$

seria geometrică

$$\sum_{m=0}^{\infty} a_m (z-z_0)^m = S(z) \quad \text{in } U(z_0; R_1)$$

$$\sum_{m=0}^{\infty} b_m (z-z_0)^m = T(z) \quad \text{in } U(z_0; R_2)$$

$$\sum_{m=0}^{\infty} (a_m+b_m)(z-z_0)^m = S(z)+T(z) \quad \text{in } U(z_0; \min\{R_1, R_2\})$$

$$\sum_{m=0}^{\infty} c\, a_m (z-z_0)^m = c\, S(z) \quad \text{in } U(z_0; R_1) \quad , c \in \mathbb{C}^*$$

$$c_m = a_m b_0 + a_{m-1} b_1 + \dots + a_0 b_m$$

$$\sum_{m=0}^{\infty} c_m (z-z_0)^m = S(z)T(z) \quad \text{in } U(z_0; \min\{R_1; R_2\})$$

4c,d) $\cos z = \dfrac{e^{iz}+e^{-iz}}{2}$, $\sin z = \dfrac{e^{iz}-e^{-iz}}{2i}$

$$e^z = 1 + \frac{1}{1!}z + \frac{1}{2!}z^2 + \dots + \frac{1}{m!}z^m + \dots \quad , z \in \mathbb{C}$$

$$e^{iz} = 1 + \cancel{\frac{1}{1!}(iz)} + \frac{1}{2!}(iz)^2 + \cancel{\frac{1}{3!}(iz)^3} + \frac{1}{4!}(iz)^4 + \cancel{\frac{1}{5!}(iz)^5} + \frac{1}{6!}(iz)^6 + \cancel{\frac{1}{7!}(iz)^7} + \dots \quad z \in \mathbb{C}$$

(+)

$$e^{-iz} = 1 + \cancel{\frac{1}{1!}(-iz)} + \frac{1}{2!}(-iz)^2 + \cancel{\frac{1}{3!}(-iz)^3} + \frac{1}{4!}(-iz)^4 + \cancel{\frac{1}{5!}(-iz)^5} + \frac{1}{6!}(-iz)^6 + \cancel{\frac{1}{7!}(-iz)^7} + \dots \quad z \in \mathbb{C}$$

$/ : 2$ $\quad i^2 = -1 \quad i^4 = 1 \quad i^6 = -1$

$$\boxed{\cos z = 1 - \frac{1}{2!}z^2 + \frac{1}{4!}z^4 - \frac{1}{6!}z^6 + \dots \quad , z \in \mathbb{C}}$$

$$e^{iz} = \cancel{1} + \frac{1}{1!}(iz) + \cancel{\frac{1}{2!}(iz)^2} + \frac{1}{3!}(iz)^3 + \cancel{\frac{1}{4!}(iz)^4} + \frac{1}{5!}(iz)^5 + \cancel{\frac{1}{6!}(iz)^6} + \frac{1}{7!}(iz)^7 + \dots \quad z \in \mathbb{C}$$

(−)

$$e^{-iz} = \cancel{1} + \frac{1}{1!}(-iz) + \cancel{\frac{1}{2!}(-iz)^2} + \frac{1}{3!}(-iz)^3 + \cancel{\frac{1}{4!}(-iz)^4} + \frac{1}{5!}(-iz)^5 + \cancel{\frac{1}{6!}(-iz)^6} + \frac{1}{7!}(-iz)^7 + \dots \quad z \in \mathbb{C}$$

$/ : (2i)$ $\quad i = i \quad i^3 = -i \quad i^5 = i \quad i^7 = -i$

$$\boxed{\sin z = \frac{1}{1!}z - \frac{1}{3!}z^3 + \frac{1}{5!}z^5 - \frac{1}{7!}z^7 + \dots \quad , z \in \mathbb{C}}$$

$$\cos z = \sum_{m=0}^{\infty} (-1)^m \frac{1}{(2m)!} z^{2m} \quad , z \in \mathbb{C}$$

$$\sin z = \sum_{m=0}^{\infty} (-1) \frac{1}{(2m+1)!} z^{2m+1} \quad , z \in C$$

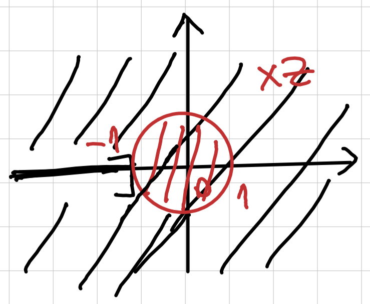
4e) $f(z) = \log(1+z)$, $\log w = \ln|w| + i \arg w$

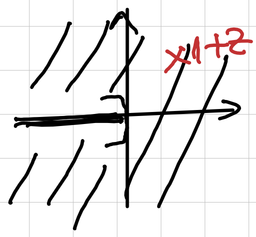
$\log 1 = \underbrace{\ln|1|}_{0} + i\underbrace{\arg(1)}_{0} = 0$

$1+z \in \mathbb{C}^-$

$z_0 = 0$

$f \in \mathcal{H}(U(0;1))$





$\mathbb{C}^-$

$\log \in \mathcal{H}(\mathbb{C}^-)$

$\mathbb{C}^- = \mathbb{C} \setminus (-\infty, 0]$

T. dezv. Taylor

$\Downarrow$

$\exists! \sum_{n=0}^{\infty} a_n z^n = \log(1+z)$, $z \in U(0;1)$

Cauchy-Hadamard d) $f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$

$(\log(1+z))' = \frac{1}{1+z}$

$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n \qquad z \in U(0;1) \qquad z \to -z$

$\frac{1}{1+z} = \sum_{n=0}^{\infty} (-z)^n = \sum_{n=0}^{\infty} (-1)^n z^n$, $z \in U(0;1)$

$\frac{1}{1+z} = \sum_{n=1}^{\infty} n a_n z^{n-1} = \sum_{n=0}^{\infty} (-1)^n z^n = \sum_{n=1}^{\infty} (-1)^{n-1} z^{n-1}$

$\underset{\text{id. coef}}{\Rightarrow} n a_n = (-1)^{n-1} \Rightarrow a_n = \frac{(-1)^{n-1}}{n}$, $n \geq 1$

$\log(1+z) = a_0 + a_1 z + a_2 z^2 + \dots \qquad z \to 0$

$\log 1 = a_0 \Rightarrow a_0 = 0$

$\log(1+z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} z^n = z - \frac{1}{2}z^2 + \frac{1}{3}z^3 - \frac{1}{4}z^4 + \dots$, $z \in U(0;1)$

$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$, $z \in U \qquad z \to z-1$

$\frac{1}{1-(z-1)} = \boxed{\frac{1}{2-z} = \sum_{n=0}^{\infty} (z-1)^n, \ |z-1| < 1}$ $\uparrow z_0 = 1$

$|z-1| < 1 \Leftrightarrow z \in U(1;1)$

⑤ a) $f(z) = \frac{z^2 - 2z + 5}{(z-2)(z^2+1)} = \underset{\substack{\uparrow \\ \text{dezv}}}{\frac{A}{z-2}} + \underset{\substack{\uparrow \\ \text{dezv}}}{\frac{B}{z-i}} + \underset{\substack{\uparrow \\ \text{dezv}}}{\frac{C}{z+i}}$

$\frac{1}{z+i} = \frac{1}{i(-iz+1)} = \frac{1}{i} \frac{1}{1-iz} = \frac{1}{i} \sum_{n=0}^{\infty} (iz)^n$, $|z| < 1$

$|i| = 1$

b) $z + z^2$

$\frac{1}{(1-z)^3} \to$ derivare, pornind de la $\frac{1}{1-z}$